

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week4 – Monday	Batch	23CSBTB47B
Name	Aelishala Manoj	Hall Ticket No	2303A54054
Assignment Number: 7.1(Present assignment number)/24(Total number of assignments)			
Q.No.	Question	Expected Time to complete	
1	Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs Lab Objectives: - To identify and correct syntax, logic, and runtime errors in Python programs using AI tools. - To understand common programming bugs and AI-assisted debugging suggestions. - To evaluate how AI explains, detects, and fixes different types of coding errors. - To build confidence in using AI to perform structured debugging practices. Lab Outcomes (LOs): After completing this lab, students will be able to: - Use AI tools to detect and correct syntax, logic, and runtime errors. - Interpret AI-suggested bug fixes and explanations. - Apply systematic debugging strategies supported by AI-generated insights. - Refactor buggy code using responsible and reliable programming patterns.	Week4 - Monday	

Task Description #1 (Syntax Errors – Missing Parentheses in Print Statement)

Task: Provide a Python snippet with a missing parenthesis in a print statement (e.g., `print "Hello"`). Use AI to detect and fix the syntax error.

Bug: Missing parentheses in print statement

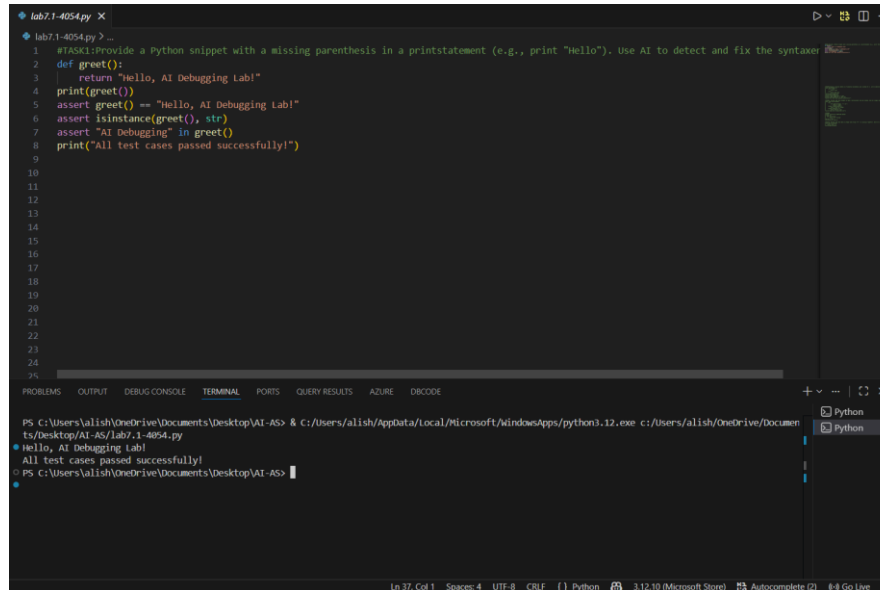
```
def greet():  
    print "Hello, AI Debugging Lab!"  
greet()
```

Requirements:

- Run the given code to observe the error.
- Apply AI suggestions to correct the syntax.
- Use at least 3 assert test cases to confirm the corrected code works.

Expected Output #1:

- Corrected code with proper syntax and AI explanation.



```
1 #task: provide a Python snippet with a missing parenthesis in a printstatement (e.g., print "Hello"). Use AI to detect and fix the syntax error.
2 def greet():
3     return "Hello, AI Debugging Lab!"
4     print(greet())
5     assert greet() == "Hello, AI Debugging Lab!"
6     assert isinstance(greet(), str)
7     assert "AI Debugging" in greet()
8     print("All test cases passed successfully!")
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```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE DECODE

```
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & c:\Users\alish\AppData\Local\Microsoft\WindowsApps\python3.12.exe c:\Users\alish\OneDrive\Documents\Desktop\AI-AS\lab7.1-4054.py
Hello, AI Debugging Lab!
All test cases passed successfully!
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS>
```

Ln 37, Col 1 Spaces: 4 UTF-8 CRLF Python 3.12.10 (Microsoft Store) Autocomplete (2) 8-8 Go Live

Task Description #2 (Incorrect condition in an If Statement)

Task: Supply a function where an if-condition mistakenly uses `=` instead of `==`. Let AI identify and fix the issue.

Bug: Using assignment (`=`) instead of comparison (`==`)

```
def check_number(n):
```

```
    if n = 10:
```

```
        return "Ten"
```

```
    else:
```

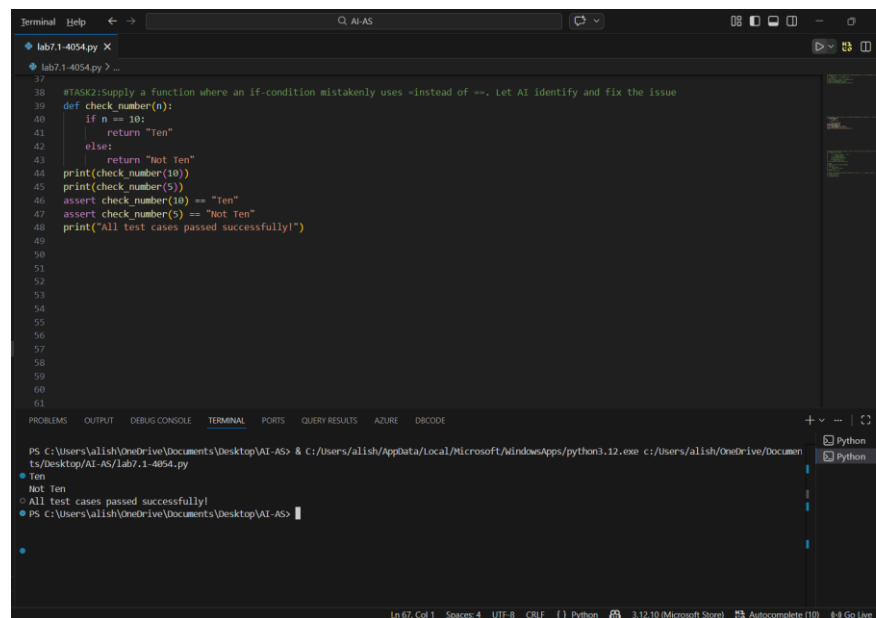
```
        return "Not Ten"
```

Requirements:

- Ask AI to explain why this causes a bug.
- Correct the code and verify with 3 assert test cases.

Expected Output #2:

- Corrected code using == with explanation and successful test execution.



```
37
38 #TASK2: Supply a function where an if-condition mistakenly uses = instead of ==. Let AI identify and fix the issue
39 def check_number(n):
40     if n = 10:
41         return "Ten"
42     else:
43         return "Not Ten"
44     print(check_number(10))
45     print(check_number(5))
46     assert check_number(10) == "Ten"
47     assert check_number(5) == "Not Ten"
48     print("All test cases passed successfully!")
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```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE DECODE

```
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & c:\users\alish\AppData\Local\Microsoft\WindowsApps\python3.12.exe c:\users\alish\OneDrive\Documents\Desktop\AI-AS\lab7.1-4054.py
Ten
Not Ten
All test cases passed successfully!
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS>
```

Ln 67, Col 1 Spaces: 4 UTF-8 CRLF Python 3.12.10 (Microsoft Store) Autocomplete (10) Go Live

Task Description #3 (Runtime Error – File Not Found)

Task: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling.

Bug: Program crashes if file is missing

```
def read_file(filename):
```

```
    with open(filename, 'r') as f:
```

```
        return f.read()
```

```
print(read_file("nonexistent.txt"))
```

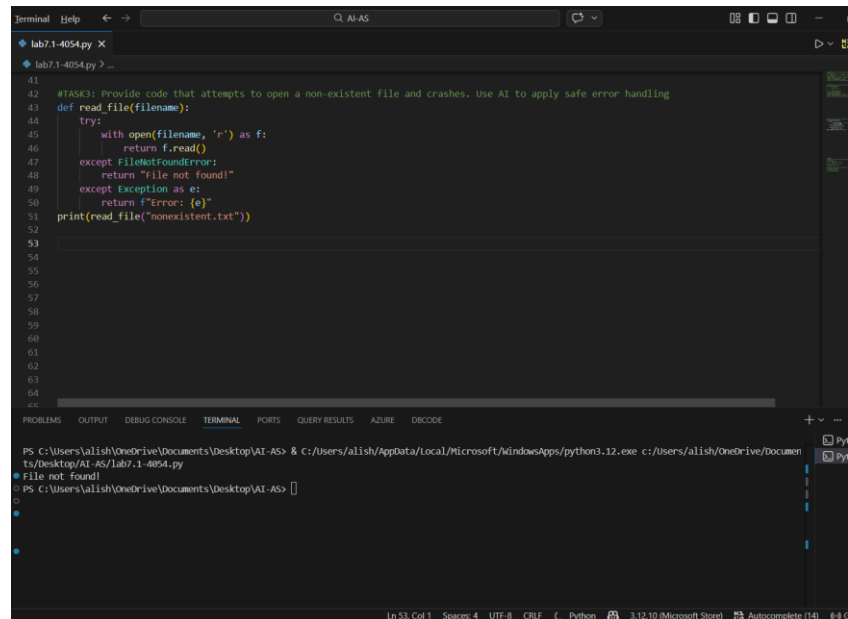
Requirements:

- Implement a try-except block suggested by AI.
- Add a user-friendly error message.
- Test with at least 3 scenarios: file exists, file missing, invalid path.

Expected Output #3:

- Safe file handling with exception management.

```
def read_file(filename):  
    try:  
        with open(filename, 'r') as f:  
            return f.read()  
    except FileNotFoundError:  
        return "File not found!"  
    except Exception as e:  
        return f"Error: {e}"  
print(read_file("nonexistent.txt"))
```



The screenshot shows a VS Code editor window with a Python file named 'lab7.1-4054.py'. The code defines a function 'read_file' that attempts to open a file and read its contents. It uses a try-except block to handle 'FileNotFoundError' and other exceptions. The function is then called with 'nonexistent.txt'. The terminal output shows the execution of the script, which results in 'File not found!'.

```
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42 #TASK3: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling  
43 def read_file(filename):  
44     try:  
45         with open(filename, 'r') as f:  
46             return f.read()  
47     except FileNotFoundError:  
48         return "File not found!"  
49     except Exception as e:  
50         return f"Error: {e}"  
51 print(read_file("nonexistent.txt"))  
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```

Task Description #4 (Calling a Non-Existent Method)

Task: Give a class where a non-existent method is called (e.g.,

obj.undefined_method()). Use AI to debug and fix.

Bug: Calling an undefined method

```
class Car:
    def start(self):
        return "Car started"

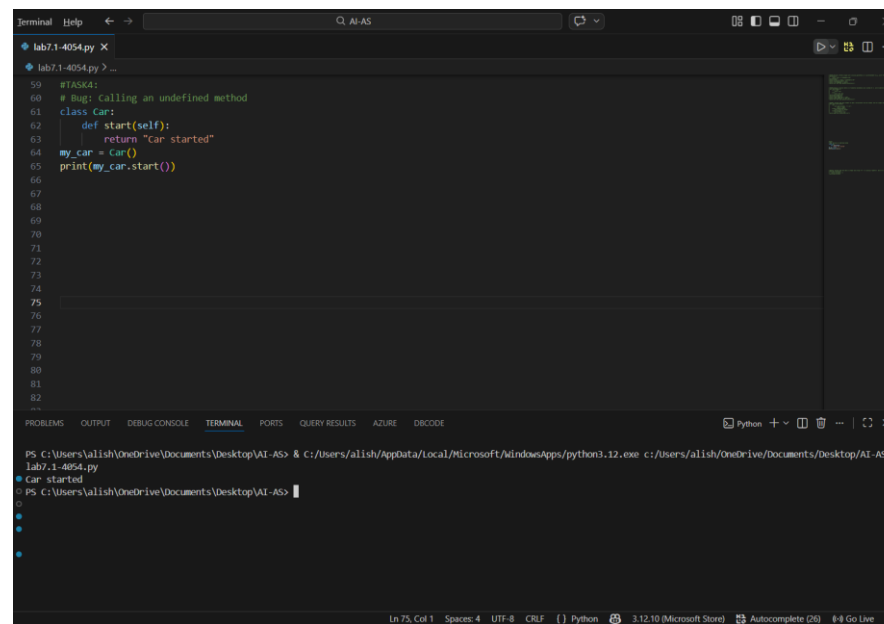
my_car = Car()
print(my_car.drive()) # drive() is not defined
```

Requirements:

- Students must analyze whether to define the missing method or correct the method call.
- Use 3 assert tests to confirm the corrected class works.

Expected Output #4:

- Corrected class with clear AI explanation.



```
lab7.1-4054.py X
lab7.1-4054.py > ...
59 # TASK4:
60 # Bug: Calling an undefined method
61 class Car:
62     def start(self):
63         return "car started"
64 my_car = Car()
65 print(my_car.start())
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE DIBCODE
Python + - - - - -
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & c:/Users/alish/AppData/Local/Microsoft/WindowsApps/python3.12.exe c:/Users/alish/OneDrive/Documents/Desktop/AI-AS/lab7.1-4054.py
Car started
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS>
Ln 75, Col 1 Spaces 4 UTF-8 CRLF Python 3.12.10 (Microsoft Store) Autocomplete (26) 44 Go Live
```

Task Description #5 (TypeError – Mixing Strings and Integers in Addition)

Task: Provide code that adds an integer and string ("5" + 2) causing a TypeError. Use AI to resolve the bug.

Bug: TypeError due to mixing string and integer

```
def add_five(value):
```

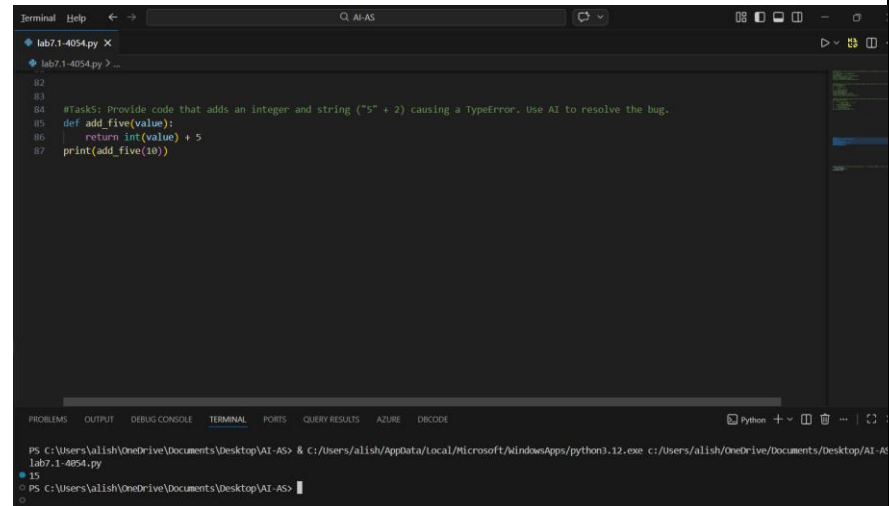
```
    return value + 5  
    print(add_five("10"))
```

Requirements:

- Ask AI for two solutions: type casting and string concatenation.
- Validate with 3 assert test cases.

Expected Output #5:

- Corrected code that runs successfully for multiple inputs.



```
lab7.1-4054.py X  
lab7.1-4054.py > ...  
82  
83  
84 #task5: Provide code that adds an integer and string ("9" + 2) causing a TypeError. Use AI to resolve the bug.  
85 def add_five(value):  
86     return int(value) + 5  
87 print(add_five(10))  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE DECODE  
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PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & c:/Users/alish/AppData/Local/Microsoft/WindowsApps/python3.12.exe c:/Users/alish/OneDrive/Documents/Desktop/AI-AS/lab7.1-4054.py  
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PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS>
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Z